

Anvit Verma

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Education

Bachelor of Technology in Computer Science
Vellore Institute of Technology

Sept 2023 – May 2027

Experience

Research Assistant

Vellore Institute of Technology

Bhopal, IN

Sept 2025 – Present

- Analyzed findings from **80+ peer-reviewed papers** to evaluate the evolution of **neural architectures** in autonomous driving from **Kalman filter** baselines to modern **LSTM** and **GNN** models.
- Determined a **40% reduction** in trajectory prediction error across modern safety-critical systems by validating reported **neural network** benchmarks.
- Appraised **9 foundational ADAS** surveys to identify research gaps and formulate a technical roadmap for **Neuro-Symbolic** and **Bio-Inspired** certifiable **AI** systems.

Technical Assistant

Official University E-Learning Portal (Vityarthi)

Bhopal, IN

March 2025 – August 2025

- Spearheaded the end-to-end technical development of the university wide **Linux** course (**CSE0002**) for **4,100+ students** in just first semester of the launch.
- Systematized **90 Bash** and **Shell** automation scripts replicating real-world system administration workflows including log analysis and process monitoring.
- Delivered **100%** of technical screen demonstrations capturing real-time execution of advanced **Linux** commands for asynchronous learning at scale.

AI Intern

Jio Platforms Limited (JPL)

Navi Mumbai, IN

May 2025 – June 2025

- Constructed autonomous **AI agents** using **LangChain** and **LLaMA 3**, implementing a **RAG pipeline** over **500+ internal documents** for precise information retrieval.
- Implemented dynamic **SQL generation** enabling structured querying across complex relational databases within the agent workflow.
- Launched a **LangGraph**-based multi-agent system that automated **90%** of complaint routing and reduced operational latency by **20%**.

Publications

Perception Challenges in Autonomous Vehicles, International Journal 2025 Publications -
Volume 1 - Issue 3, Airo International Research Journal ISSN 2320-3714
Ishaan Shrivastava, **Anvit Verma**

Mar 2025

[\[Link\]](#)

Projects

Causal Evaluation of Maharashtra EV Policy

[\[GitHub\]](#)

Python, DuckDB, Polars, EconML (Causal Forests), Scipy, Statsmodels

- Engineered a memory-efficient, out-of-core data engineering pipeline using **DuckDB** and **Polars** to ingest, join, and lazily evaluate over 5 million API-sourced vehicle registrations and hourly PM2.5 readings into a balanced 16-district monthly panel dataset.
- Pioneered a rigorous quasi-experimental causal analysis utilizing the **Synthetic Control Method (SCM)** and **Double Machine Learning**, mathematically validating a +14.3% surge in EV adoption ($p < 0.001$ via in-space placebo permutation tests) and extracting actionable Heterogeneous Treatment Effects using **SHAP** values.

GitHub Repo Analyst AI

[\[Live Demo\]](#) [\[GitHub\]](#)

LangGraph, LLaMA 3, ChromaDB, React (Vite), Framer Motion

- Architected an autonomous **LangGraph** orchestration layer with **LLaMA 3 agents** and a **ChromaDB RAG** pipeline, enabling conversational codebase analysis, repository discovery, and automated README generation.
- Engineered an interactive **React** frontend with a flawless **Windows XP-themed desktop experience**, completely automating complex **three-stage security audits** (SAST & dependencies) directly within the OS simulation.

Tastelytics – Serverless Music Review API

AWS Lambda, DynamoDB, API Gateway, Cognito, CodePipeline

- Architected a scalable serverless backend for a dynamic **music review platform** using **AWS Lambda** and **API Gateway**, leveraging **DynamoDB Global Secondary Indexes (GSI)** to deliver **sub-300ms** queries for community interactions.
- Secured role-based user authentication via **Amazon Cognito** and integrated the **Spotify API** for real-time track metadata, deployed through a **zero-downtime CI/CD** pipeline using **AWS CodePipeline**.

Skills

Languages: C++, Python, Java, JavaScript, SQL, Linux

AI & Machine Learning: SR-GANs, LLMs, LangSmith, Multi-Agent Orchestration, RAG Pipelines, Transformers, Computer Vision, Causal Machine Learning

Frameworks & Libraries: PyTorch, TensorFlow, LangChain, LangGraph, React, Node.js, Pandas, Polars, NumPy, Scikit-learn, EconML, Statsmodels, DuckDB

Cloud & Tools: AWS (Lambda, S3, DynamoDB, API Gateway, Cognito, CodePipeline), Docker, Git, CI/CD, Postman